

QuantumSentinel: A Rigorous Benchmark of Variational Quantum Neural Networks for Zero-Day and Adversarial Intrusion Detection in IoT Networks

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1. Introduction & Motivation

The rapid expansion of Internet of Things (IoT) ecosystems has significantly increased the attack surface of modern cyber infrastructure. Existing intrusion detection systems (IDS) struggle against evolving threats such as zero-day attacks, adversarial perturbations, and cross-domain attack transferability. Recent literature has proposed Variational Quantum Classifiers (VQCs) and Quantum Neural Networks (QNNs) as promising alternatives to classical IDS architectures, claiming improved detection capability and robustness.

However, most existing Quantum Machine Learning (QML) intrusion detection studies rely on three major simplifications:

- Evaluation on a single dataset
- Training on noiseless quantum simulators
- Absence of adversarial robustness analysis

QuantumSentinel aims to rigorously stress-test these claims through a unified, reproducible benchmark spanning multiple IoT datasets, realistic NISQ noise conditions, adversarial attacks, and zero-day intrusion scenarios. The project focuses not on proving quantum superiority, but on objectively evaluating where quantum models succeed, fail, and generalize compared to strong classical baselines.

2. Goals and Objectives

- Build a unified preprocessing and evaluation benchmark across CICIoT2023, TON_IoT, and UNSW-NB15.
- Implement and train three quantum architectures: hardware-efficient ansatz, strongly-entangling ansatz, and data re-uploading QNN.
- Compare angle, amplitude, and IQP encoding strategies.
- Develop parameter-matched classical baselines including Random Forest, XGBoost, CNN-LSTM, and transformer-based IDS.
- Conduct experiments on:
 - Cross-dataset generalization
 - Zero-day robustness
 - NISQ noise robustness
 - Adversarial robustness using FGSM and PGD
- Perform paired t-tests and McNemar's tests across multiple random seeds.

3. Weekly Breakdown

Week 1: Literature Audit, Setup & Protocol Design

- Environment setup using PennyLane, Qiskit, PyTorch, and Scikit-learn.
- Review 2024–2025 IoT-QML literature.
- Finalize datasets, zero-day splits, noise parameters, and adversarial budgets.

Week 2: Benchmark Construction & Preprocessing

- Unified preprocessing across CICIoT2023, TON_IoT, and UNSW-NB15.
- Feature engineering, normalization, PCA, and feature ranking.
- Construct zero-day and cross-dataset evaluation protocols.

Week 3: Encoding & Circuit Design

- Implement angle, amplitude, and IQP encodings.
- Design three ansatz circuit families.
- Conduct expressibility and entanglement diagnostics.

Week 4: Full Model Training

- Train all quantum models across datasets and encodings.
- Train parameter-matched classical baselines.
- Log reproducibility metadata and hyperparameters.

Week 5: Robustness & Hardware Experiments

- Conduct cross-dataset, zero-day, noisy, and adversarial evaluations.
- Perform FGSM and PGD attacks.
- Run one configuration on IBM Quantum free-tier hardware.

Week 6: Manuscript Preparation

- Draft Research Paper and project document.
- Prepare figures, tables, and reproducibility statement.
- Record demonstration and presentation materials.

Final Stretch (12 August – 15 August)

- Final repository cleanup.
- Dockerfile and README completion.
- Final presentation and manuscript handoff.

4. Guidance & Communication

- Weekly mentor review meetings.
- Parallel sub-team coordination for Data, Quantum Circuits, and Evaluation.
- Continuous GitHub code reviews and experiment tracking.
- Shared documentation via Notion and Google Drive.

5. Tools & Resources

- Python, PyTorch, Scikit-learn, XGBoost
- Qiskit and PennyLane

- IBM Quantum hardware and PennyLane Lightning simulator
- Docker, GitHub Actions
- Datasets: CICIoT2023, TON_IoT, UNSW-NB15

6. Expected Deliverables / Outcomes

- Reusable benchmark framework for IoT-QML intrusion detection.
- Three trained quantum IDS architectures across multiple encoding strategies.
- Parameter-matched classical baselines.
- Complete evaluation across in-distribution, zero-day, noisy, and adversarial settings.
- Statistical significance analysis across all comparisons.
- Public reproducibility repository with code, weights, logs, Dockerfile, and rerun scripts.